METABOLIC STABILITY IN LIVER MICROSOMES

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
%	Percentage
°C	Degree Celsius
µL	Microlitre
µM	Micromolar
CL	Clearance
COA	Certificate of analysis
conc^{n}	Concentration
CYP	Cytochrome P450
DMPK	Drug metabolism and pharmakokinetics
DMSO	Dimethyl sulfoxide
g	Gram
HLM	Human liver microsomes
HOD	Head of the department
HPLC	High performance liquid chromatography
kg	Kilogram
LC-MS/MS	Liquid chromatography coupled with tandem mass spectrometry
LNB	Lab notebook
mg	Milligram
min	Minutes
mL	Millilitre
MLM	Mouse liver microsomes
mM	Millimolar
MS	Microsoft
N	Normality
NA	Not applicable
NADPH	Nicotinamide adenine dinucleotide phosphate hydrogen
PD	Pharmaceutical development
pH	Negative logarithm to base 10 of the positive hydrogen ion molar concentration
PPE	Personal protective equipment
RLM	Rat liver microsomes
rpm	Revolutions per minute
S. No.	Serial number

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1. OBJECTIVE

The objective of this experiment is to determine the metabolic stability of test compound(s) in liver microsomes.

2. SCOPE

The procedure described in this protocol will be used for assessment of metabolic stability of test compound(s) in rat, mouse, mini pig, dog and/ or human liver microsomes.

3. RESPONSIBILITY

All personnel performing this study are responsible to follow the process outlined in this protocol, at Aurigene Pharmaceutical Services Ltd, India.

4. INTRODUCTION

Metabolic stability is defined as the amount of substrate metabolized upon incubation with liver microsomes. Liver microsomes are the sub-cellular particles derived from endoplasmic reticulum of hepatic cells. Microsomes are a rich source of drug metabolizing enzymes, including CYP, which play a key role in the metabolism of a wide range of xenobiotics. Hence this assay is an important in vitro tool which is often predictive of the hepatic clearance in vivo. Stability data of a compound in microsomes from different species is critical in selecting the species for preclinical studies and ascertaining the dose projections for administration in humans.

5. SAFETY AND PRECAUTIONS

During the conduct of the study, the study personnel should take necessary safety precautions and use personal protective equipment (PPE) such as facemasks, gloves, lab coat, goggles, shoes etc.

DMSO:

- Keep away from sources of ignition.
- Do not breathe vapour or mist.
- Avoid contact with skin, eyes and clothing.

6. MATERIALS AND EQUIPMENT

6.1 EQUIPMENT

Equipment	**Make**
Analytical balance	Sartorius/ Denver
Conical glass vials	Thermoscientific, Chromacol 1.1 STVG
Glass Pasteur pipettes	VWR, 612-1701
HPLC	Agilent/ Shimadzu
Micro centrifuge tubes	Tarsons/ Phenomenex
Micro-titre plate shaker	IKA® MTS
pH meter	Horiba scientific
Pipettes (10, 100, 200 and 1000 µL)	Eppendorf
Refrigerator	LG
Software	Microsoft excel

Equipment	Make
Sonicator	Shalom instruments
Vortex	IKA®Vortex
Vortex mixer	Neuation; *i Swix MV*

Note:

- Details of all equipment/ instruments used during the study should be provided in the LNB & report (if any).

6.2 CHEMICALS

Chemicals	Make; Catalogue No.
Acetonitrile	J.T.Baker; 9012-03
Bosutinib	Sigma Aldrich; PZ0192
Dimethyl sulfoxide	Sigma Aldrich, D8418
Glyburide	Sigma Aldrich; G2539
NADPH	SRL; 99197
Phosphate buffer powder	Sigma; P7994

Note:

- Complete list of chemicals used, and their details should be provided in the LNB & report (If any).
- Liver microsomes (of all species) can be procured from certified vendors with COA.
- Equivalent grade chemicals/ reagents from alternate sources can be used on need basis.
- Source information of microsomes should be provided along with results.

6.3 PREPARATION OF SOLUTIONS AND STORAGE CONDITIONS

S. No.	Composition	Storage condition
1.	Preparation of 100 mM phosphate buffer, pH 7.4 Reconstitute the entire contents of the bottle (Sigma; P7994) in 3.8 L of water to prepare a 100 mM solution, pH 7.4 at 25°C	2-8°C 3 months
2.	Preparation of stop solution Prepare 1 mg/mL stock solutions of the internal standard (Glyburide) in DMSO and dilute with acetonitrile to obtain 250ng/mL final concentration.	2-8°C
3.	Preparation of microsomal working solution Prepare 2.0 mg/mL working solution of microsomal protein from 20 mg/mL stock in 100 mM phosphate buffer.	Prepare fresh (store on ice until use)
4.	Preparation of test/ reference compound(s) Prepare 10 mM primary stock solution in DMSO. Dilute the primary stock solution in DMSO to obtain 1 mM intermediate working	Stock solution: 20-25°C, 3 months (Working solutions:

S. No.	Composition	Storage condition
	solution. Dilute the intermediate working solution in buffer to obtain 4 µM working solution maintaining the organic solvent ≤ 0.1%	20-25°C, Prepare fresh)
5.	Preparation of NADPH solution Prepare suitable stock solution in de-ionized water. Dilute the stock solution in buffer to obtain 2 mM working solution.	Stock solution: -20°C Working solution: Prepare fresh (store on ice until use)

Note:

- Changes to the above-mentioned solution preparation methods should be captured in the LNB & report (If any).
- Final concentration in reaction; 1) NADPH = 1 mM; 2) Microsomal protein = 0.5 mg/mL 3) Test/ reference compound = 1 µM
- The protein becomes less active due to repeated freeze-thaw of the stock vial. To prevent this, whenever a new vial of microsomes is thawed, the entire contents should be transferred to 0.5 mL centrifuge tubes in 50 µL/ 100 µL aliquots and frozen at -80°C. Only the required number of vials can be thawed for each experiment.
- Final concentration of acetonitrile in reaction should be < 2% and DMSO concentration ≤ 0.1%. Higher concentration of DMSO will potently inhibit the enzyme activity.

7. PROCEDURE

Perform the assay in singlet (n = 1).

Prepare the test and reference compound(s) dilutions as described in section 6.3 in a 96 deep well plate and mix properly.

Prepare the microsomal and NADPH working solutions as described in section 6.3.

Dispense 25 µL of test/ reference compound(s) into designated wells in a deep well plate.

Add 25 µL of microsomal working solution and mix well for 30 sec at 450 rpm.

Cover the incubation plate with silicon sealing mat and pre-incubate at 37 °C on plate shaker at 450 rpm for 15 min.

Start the reaction by addition of 50 µL of pre-incubated NADPH solution into desired wells in a deep well plate and incubate at 37°C without shaking.

At time points $T_{0.5}$, T_5, T_{15}, and T_{30} min, draw 20 µL samples and transfer to separate wells in the 96 well deep well plate containing 90 µL of ice cold stop solution and place the plate in the refrigerator until the end of the assay.

Remove the precipitated protein by centrifugation at 4000 rpm for 10 min at 4°C.

Transfer the 75 µL supernatant into a new 96-well plate and add 90 µL of milli-Q water into desired wells of 96 well plate and vortex for 3 min.

Submit for analysis by LC-MS/MS.

If not analyzed immediately, store the samples at 2-8°C.

Note:

- Any changes of the protocol should be discussed and agreed with Novartis. New changes have to be communicated and agreed upfront before uploading results. In any case such changes must be captured in LNB and report.
- Suitable bioanalytical method will be used for the analysis of the samples.

8. ASSAY CONTROL

Bosutinib should be included as assay control.

Study would be qualified based on the acceptance criteria given in the data template.

Note:

- Any changes to the assay controls would be made only with the consent of the sponsor and should be described in the LNB & report ((If any).

9. DATA RECORDING

All the raw data, chromatograms and reports should be stored in the server at Aurigene Pharmaceutical Services Limited, India for a period of five years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and the same should be archived as per the instructions/ procedure of Intellectual Property Management (IPM) department.

10. DATA ANALYSIS

Measure the initial amount of substrate concentration at $T_{0.5}$ and assess the metabolic stability by disappearance of the compound at various time points based on the change in analyte to internal standard peak area ratio.

Calculate the following parameters using the given formulae:

a) % parent remaining at time point T_x:

$$\% \text{ parent remaining} = \frac{T_x \text{ peak area ratio}}{\text{maximal peak area ratio}} \times 100$$

b) Half-life (t½): Unit = min

Plot a graph for the percent parent compound remaining at each time point versus time in min using graph pad prism. Select nonlinear one phase decay model to obtain the half-life.

c) Intrinsic clearance (CL_{int}): Unit = µL/min/mg

$$CL_{int} = \frac{\ln 2}{\text{half-life}} \times \frac{1000}{[\text{Microsomal protein conc}^n]}$$

11. REPORTING

Data should be reported in an MS excel format. Based on the need, MS word reports will may also be prepared.

If reference compound(s) does not meet the specified acceptance criteria, the complete batch should be rejected.

The rules given in the table below should be followed for reporting.

Condition	**Report Status**	**Remark**
No compound available for assay	Undefined	Empty tube

Precipitation observed when preparing compound plate	Undefined	Precipitation in DMSO stock solution
Calculated Clint<25µL/min/mg (all data points to be within 60 &- 100 % when Clint < 25 µL/min/mg)	Report Clint < 25	N/A
Calculated Clint > 700	Report Clint > 700	N/A
Test compound not detected by 5 min due to high clearance. T0 min peak area high (Agrees with optimization)	Report Clint > 700	N/A
Test compound not detected by 5 min due to high clearance. T0 min peak area low (Disagrees with optimization)	Undefined	Very high clearance likely
Clint < 25 µL/min/mg and at least one data point below 60% remaining	Undefined	Data poorly fits model used for calculating parameters
Clint = 25-50 µL/min/mg and SD% of fit is > 50%	Undefined	Data poorly fits model used for calculating parameters
Clint >50 µL/min/mg and SD % of fit is > 25%	Undefined	Data poorly fits model used for calculating parameters
Test compound not detectable	Undefined	Analytical issue - No peak detected
Double peak observed in chromatogram for assay and also appears during optimization check. Clint > factor of 2 depending upon integration chosen.	No data reported	Analytical issue - Double peak in chromatogram
Unidentified double peak observed in chromatogram for assay but not present during optimization check, question over data validity.	No data reported	Analytical issue - Double peak in chromatogram
Poor peak shape observed, and accuracy of result compromised	No data reported	Analytical issue- Poor chromatogram

12. DATA ARCHIVAL

All the raw data, chromatograms and reports should be stored in the server at Aurigene Discovery Technologies Limited, India for a period of two years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and the same should be archived as per the instructions/ procedures of Intellectual Property Management (IPM) department.

13. REFERENCES

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- Crespi CL, Stresser DM. Journal of pharmacology and toxicology methods, 44, 325-331; 2000.
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- Brian D and Tim M; Physiological parameters in laboratory animals and humans. Pharmaceutical research, Vol.10, (7); 1993.

14. CHANGE AND AMENDMENT HISTORY

Change History		
Version	**Changes made/ Reason for changes**	**Effective date**
2.0	Revised Version	01 Jul 2022